

	QATAR FOUNDATION QATAR NATIONAL LIBRARY			
  	Location EDUCATION CITY - QATAR	Contract No.	Document No.	
				
	Title: BAS SYSTEM DESCRIPTION FOR QATAR NATIONAL LIBRARY (QNL)	Page	Rev No.	
		414	0	1

3.3.8 CPNL-0008 Instrument List

SMART INSTRUMENTS LIST FOR CPNL-0008										
Sr. No.	Type of Instrument	Description	Process Fluid	Signal Type	Range	Type	Make	Model No.	Supply Voltage	Quantity
N/A										

NON-SMART INSTRUMENTS LIST FOR CPNL-0008										
Sr. No.	Type of Instrument	Description	Process Fluid	Signal Type	Range	Type	Make	Model No.	Supply Voltage	Quantity
1	Space Temperature	Roof Floor DX Unit & Zone Temperature Sensors	Air	Analog	0...50 °C	PT-100	Greystone	TSRC2C	24 VDC	4
2	Pressure Transmitter	Fire Pump Remote Pressure Transmitters	Water	Analog	0...30000kPa	Analog	Greystone	HWP-G-108-LCD	24 VAC	2

3.3.9 Unitary Controller (FCU/CAV/VAV) Instrument List

SMART INSTRUMENTS LIST FOR FCU/CAV/VAV										
Sr. No.	Type of Instrument	Description	Process Fluid	Signal Type	Range	Type	Make	Model No.	Supply Voltage	Quantity
N/A										

NON-SMART INSTRUMENTS LIST FOR FCU/CAV/VAV										
Sr. No.	Type of Instrument	Description	Process Fluid	Signal Type	Range	Type	Make	Model No.	Supply Voltage	Quantity
1	Duct Temperature	Supply Air Duct Temperature	Air	Analog	-40...100 °C	NTC-20k	MAMAC	TE-701-A-17-A	N/A	434
2	Duct Temperature	Return Air Duct Temperature	Air	Analog	-40...100 °C	NTC-20k	MAMAC	TE-701-A-17-A	N/A	287
3	Room Temperature	VAV/CAV Room Temperature	Air	Analog	0...50 °C	NTC-20k	Honeywell	CP-WM-VAV	24 VAC	12
4	Room Temperature	FCU Room Temperature	Air	Analog	0...50 °C	NTC-20k	Honeywell	CP-WM-FCU	24 VAC	44
5	Damper Actuator	VAV/CAV Supply Air Damper Actuator	Air	Analog	N/A	Modulating	Honeywell	N10010	24 VAC	297
6	Electric Valve Actuator	FCU Chilled Water Control Valve	Water	Floating	NA	Modulating	Honeywell	M6410A	24 VAC	137

	QATAR FOUNDATION QATAR NATIONAL LIBRARY					
  	Location EDUCATION CITY - QATAR	Contract No.	Document No.			
						
	Title: BAS SYSTEM DESCRIPTION FOR QATAR NATIONAL LIBRARY (QNL)	Page	Rev No.			
		415	0	1		

3.4 Valve Schedule

3.4.1 AHU & Heat Exchanger Valve Schedule

S/N	Tag No.	Location	Qty	Pipe Size (mm)	Flow Rate (L/s)	EWT/LWT (°C)	Max. Pressure Drop (kPa)	Appx pressure drop Bar	Flow (m3/h)	Root pressure drop	Cal Kv	sel Kv	Valve size	Pressure drop at sel kv Bar	Approx.Authority	Coil pd Bar	Valve Type	Actuator Model
1	Control valve - PHX/B/01	District Cooling Side	1	200	62.80	6.5/15.4	50.5	0.515	226.08	0.72	315.03	361.00	150.00	0.39	0.44	0.505	62.7	ITQ-240
2	Control valve - PHX/B/02	District Cooling Side	1	200	62.80	6.5/15.4	50.5	0.515	226.08	0.72	315.03	361.00	150.00	0.39	0.44	0.505	62.7	ITQ-240
3	Control valve - PHX/B/03	District Cooling Side	1	200	62.80	6.5/15.4	50.5	0.515	226.08	0.72	315.03	361.00	150.00	0.39	0.44	0.505	62.7	ITQ-240
4	Control valve - PHX/B/04	District Cooling Side	1	200	62.80	6.5/15.4	50.5	0.515	226.08	0.72	315.03	361.00	150.00	0.39	0.44	0.505	62.7	ITQ-240
5	Control valve - PHX/B/05	District Cooling Side	1	100	8.00	6.5/15.4	46.9	0.479	28.80	0.69	41.61	50.00	50.00	0.33	0.41	0.469	62.7	ITQ-100Q
6	AHU-B-01	Plant Room B4	1	100	9.48	6.5/15.4	45.13	0.4613	34.13	0.68	50.25		80.00				3-way valve- Type 3244	IQL-04
7	AHU-B-02	Plant Room B1	1	100	12.52	6.5/15.4	20.3	0.213	45.07	0.46	97.66	122.00	80.00	0.14	0.40	0.203	62.7	ITQ-100Q
8	AHU-B-03	Plant Room B2	1	100	12.52	6.5/15.4	20.3	0.213	45.07	0.46	97.66	122.00	80.00	0.14	0.40	0.203	62.7	ITQ-100Q
9	AHU-B-04	Plant Room B3	1	100	12.52	6.5/15.4	20.3	0.213	45.07	0.46	97.66		80.00				3-way valve- Type 3244	IQL-04
10	AHU-B-05	Plant Room B4	1	125	15.69	6.5/15.4	37.38	0.3838	56.48	0.62	91.17	95.00	80.00	0.35	0.49	0.3738	62.7	ITQ-100Q
11	AHU-B-06	Plant Room B4	1	125	15.68	6.5/15.4	37.15	0.3815	56.45	0.62	91.39	95.00	80.00	0.35	0.49	0.3715	62.7	ITQ-100Q
12	AHU-B-07	Plant Room B4	1	125	15.68	6.5/15.4	37.15	0.3815	56.45	0.62	91.39	95.00	80.00	0.35	0.49	0.3715	62.7	ITQ-100Q
13	AHU-B-08	Plant Room B4	1	100	9.04	6.5/15.4	47.1	0.481	32.54	0.69	46.92	58.00	50.00	0.31	0.40	0.471	62.7	ITQ-100Q
14	AHU-B-09	Plant Room B3	1	100	12.05	6.5/15.4	18.75	0.1975	43.38	0.44	97.61	122.00	80.00	0.13	0.40	0.1875	62.7	ITQ-100Q
15	AHU-B-10	Plant Room B2	1	80	5.6	6.5/15.4	25.84	0.2684	20.16	0.52	38.91		50.00				3-way valve- Type 3244	IQL-04
16	AHU-B-11	Plant Room B1	1	100	11.85	6.5/15.4	27.66	0.2866	42.66	0.54	79.69		80.00				3-way valve- Type 3244	IQL-04
17	AHU-B-12	Plant Room B1	1	100	6.29	6.5/15.4	28	0.29	22.74	0.54	42.23	50.00	50.00	0.21	0.42	0.28	62.7	ITQ-100Q
18	AHU-B-13	Plant Room B2	1	100	6.81	6.5/15.4	28.38	0.2938	25.20	0.54	46.49	58.00	50.00	0.19	0.40	0.2838	62.7	ITQ-100Q
19	AHU-B-14	Plant Room B3	1	100	9.33	6.5/15.4	26.44	0.2744	33.59	0.52	64.12	71.00	100.00	0.22	0.46	0.2644	62.7	ITQ-100Q
20	AHU-B-15	Plant Room B3	1	125	15.47	6.5/15.4	42.78	0.4378	55.69	0.66	84.17	92.00	100.00	0.37	0.46	0.4278	62.7	ITQ-100Q

	QATAR FOUNDATION QATAR NATIONAL LIBRARY					
  	Location EDUCATION CITY - QATAR	Contract No.	Document No.			
						
	Title: BAS SYSTEM DESCRIPTION FOR QATAR NATIONAL LIBRARY (QNL)	Page	Rev No.			
		416	0	1		

3.4.2 FCU Valve Schedule

Equipment Name		Area Served	Valve Flow Rate		Pressure Drop (kPa)		Total Pressure Drop		Cv Value		Valve Pd	Authority %	Pipe Size (mm)		Valve Type	Actuator Model
			LPS	GPM	Coil	Fittings	kPa	psi	Required	Selected			Actual	Selected		
1	FCU/B/01	Basement Zone 03 IDF Room B155	0.36	5.71	250.00	25.00	275.00	39.89	0.90	0.70	66.45	62.49	25.00	15.00	V5862A2039	M7410C1007
2	FCU/B/02	Basement Zone 01 Freezer Room	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
3	FCU/B/03	Basement Zone 01 IDF (Corridor) B062	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
4	FCU/B/04	Basement Zone 04 IDF (FM Storage) B215	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
5	FCU/B/05	Basement Zone 04 HL Plant Room B140	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
6	FCU/B/06	Basement Zone 01 Fire Command Room B071	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
7	FCU/B/07	Basement Zone 01 Sprinkler Pump Room B072	0.36	5.71	249.00	24.90	273.90	39.73	0.91	0.70	66.45	62.58	25.00	15.00	V5862A2039	M7410C1007
8	FCU/B/08	Basement Zone 01 Gas Suppression Room B087	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
9	FCU/B/09	Basement Zone 01 People Mover Room B088	0.27	4.28	146.00	14.60	160.60	23.29	0.89	0.70	37.38	61.61	25.00	15.00	V5862A2039	M7410C1007
10	FCU/B/10	Basement Zone 02 Security Control Room B105	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
11	FCU/B/11	Basement Zone 01 Plant Room 01 B063	0.21	3.33	300.00	30.00	330.00	47.86	0.48	0.50	44.32	48.08	20.00	15.00	V5862A2021	M7410C1007
12	FCU/B/12	Basement Zone 01 Plant Room 01 B063	0.21	3.33	300.00	30.00	330.00	47.86	0.48	0.50	44.32	48.08	20.00	15.00	V5862A2021	M7410C1007
13	FCU/B/13	Basement Zone 01 Plant Room 02 B070	0.16	2.54	182.00	18.20	200.20	29.04	0.47	0.50	25.73	46.98	20.00	15.00	V5862A2021	M7410C1007
14	FCU/B/14	Basement Zone 01 Plant Room 02 B070	0.16	2.54	182.00	18.20	200.20	29.04	0.47	0.50	25.73	46.98	20.00	15.00	V5862A2021	M7410C1007
15	FCU/B/15	Basement Zone 02 Plant Room 03 B110	0.24	3.80	382.00	38.20	420.20	60.94	0.49	0.50	57.88	48.71	20.00	15.00	V5862A2021	M7410C1007
16	FCU/B/16	Basement Zone 02 Plant Room 03 B110	0.24	3.80	382.00	38.20	420.20	60.94	0.49	0.50	57.88	48.71	20.00	15.00	V5862A2021	M7410C1007
17	FCU/B/17	Basement Zone 02 Plant Room 03 B110	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
18	FCU/B/18	Basement Zone 04 Plant Room 04 B220	0.21	3.33	300.00	30.00	330.00	47.86	0.48	0.50	44.32	48.08	20.00	15.00	V5862A2021	M7410C1007
19	FCU/B/19	Basement Zone 04 Plant Room 04 B220	0.21	3.33	300.00	30.00	330.00	47.86	0.48	0.50	44.32	48.08	20.00	15.00	V5862A2021	M7410C1007
20	FCU/B/20	Basement Zone 04 Plant Room 04 B220	0.21	3.33	300.00	30.00	330.00	47.86	0.48	0.50	44.32	48.08	20.00	15.00	V5862A2021	M7410C1007
21	FCU/B/21	Basement Zone 04 Plant Room 04 B220	0.21	3.33	300.00	30.00	330.00	47.86	0.48	0.50	44.32	48.08	20.00	15.00	V5862A2021	M7410C1007
22	FCU/B/22	Basement Zone 04 FM Storage B210	0.15	2.38	162.00	16.20	178.20	25.85	0.47	0.50	22.61	46.66	20.00	15.00	V5862A2021	M7410C1007
23	FCU/B/23	Basement Zone 04 FM Storage B210	0.15	2.38	162.00	16.20	178.20	25.85	0.47	0.50	22.61	46.66	20.00	15.00	V5862A2021	M7410C1007
24	FCU/B/24	Basement Zone 04 PH Plant Room B211	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
25	FCU/B/25	Basement Zone 04 Misting Pump Room B212	0.55	8.72	134.00	13.40	147.40	21.38	1.89	1.90	21.05	49.62	32.00	15.00	V5862A2054	M7410C1007
26	FCU/B/26	Basement Zone 04 Misting Pump Room B212	0.55	8.72	134.00	13.40	147.40	21.38	1.89	1.90	21.05	49.62	32.00	15.00	V5862A2054	M7410C1007
27	FCU/B/27	Basement Zone 01 Sorting Shelving Room B121	0.05	0.79	86.00	8.60	94.60	13.72	0.21	0.20	15.70	53.37	15.00	15.00	V5862A2005	M7410C1007
28	FCU/B/28	Basement Zone 01 Sorting Shelving Room B121	0.05	0.79	86.00	8.60	94.60	13.72	0.21	0.20	15.70	53.37	15.00	15.00	V5862A2005	M7410C1007
29	FCU/B/29	Basement Zone 03 Vestibule (Heritage Collection Entrance Area)	0.22	3.49	330.00	33.00	363.00	52.65	0.48	0.50	48.64	48.02	20.00	15.00	V5862A2021	M7410C1007
30	FCU/B/30	Basement Zone 04 Water Feature Room B226	0.51	8.08	115.00	11.50	126.50	18.35	1.89	1.90	18.10	49.66	32.00	15.00	V5862A2054	M7410C1007
31	FCU/B/31	Basement Zone 01 IDF Room B215A	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
32	FCU/B/32	Basement Zone 01 Shell Space B132	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007

		QATAR FOUNDATION QATAR NATIONAL LIBRARY						
  		Location EDUCATION CITY - QATAR	Contract No.		Document No.			
								
		Title: BAS SYSTEM DESCRIPTION FOR QATAR NATIONAL LIBRARY (QNL)	Page		Rev No.			
			417	0	1			

Equipment Name		Area Served	Valve Flow Rate		Pressure Drop (kPa)		Total Pressure Drop		Cv Value		Valve Pd	Authority %	Pipe Size (mm)		Valve Type	Actuator Model
			LPS	GPM	Coil	Fittings	kPa	psi	Required	Selected			Actual	Selected		
33	FCU/B/33	Basement Zone 01 Shell Space B132	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
34	FCU/B/34	Basement Zone 01 Shell Space B132	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
35	FCU/B/35	Basement Zone 01 Shell Space B132	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
36	FCU/B/36	Basement Zone 01 IT Office B122	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
37	FCU/B/37	Basement Zone 01 Digitization Server Space B200	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
38	FCU/B/38	Basement Zone 03 Electrical Room	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
39	FCU/1F/01	Terrace Floor Zone 3 South-East IDF Room	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
40	FCU/1F/02	First Floor Zone 01 ICT Room 02 L1030	0.36	5.71	250.00	25.00	275.00	39.89	0.90	0.70	66.45	62.49	25.00	15.00	V5862A2039	M7410C1007
41	FCU/1F/03	First Floor Zone 01 IDF Room 02 L1099	0.36	5.71	250.00	25.00	275.00	39.89	0.90	0.70	66.45	62.49	25.00	15.00	V5862A2039	M7410C1007
42	FCU/1F/04	First Floor Zone 03 Bridge Ceiling Void BR6	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
43	FCU/1F/05	First Floor Zone 03 Bridge Ceiling Void BR6	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
44	FCU/1F/06	First Floor Zone 03 Bridge Ceiling Void BR6	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
45	FCU/1F/07	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
46	FCU/1F/08	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
47	FCU/1F/09	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
48	FCU/1F/10	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
49	FCU/1F/11	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
50	FCU/1F/12	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
51	FCU/1F/13	First Floor Zone 03 Bridge Ceiling Void BR6	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
52	FCU/1F/14	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
53	FCU/1F/15	First Floor Zone 03 Bridge Ceiling Void BR6	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
54	FCU/1F/16	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
55	FCU/1F/17	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
56	FCU/1F/18	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
57	FCU/1F/19	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
58	FCU/1F/20	First Floor Zone 03 Bridge Ceiling Void BR3	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
59	FCU/1F/21	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
60	FCU/1F/22	First Floor Zone 03 Bridge Ceiling Void BR3	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
61	FCU/1F/23	First Floor Zone 03 Bridge Ceiling Void BR3	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
62	FCU/1F/24	First Floor Zone 03 Bridge Ceiling Void BR3	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
63	FCU/1F/25	First Floor Zone 03 Bridge Ceiling Void BR2	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
64	FCU/1F/26	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
65	FCU/1F/27	First Floor Zone 02 Cafe L1141	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
66	FCU/1F/28	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
67	FCU/1F/29	First Floor Zone 02 Cafe L1141	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007

		QATAR FOUNDATION QATAR NATIONAL LIBRARY					
  		Contract No.		Document No.			
		Location EDUCATION CITY - QATAR					
		Page		Rev No.			
		418		0	1		

Equipment Name		Area Served	Valve Flow Rate		Pressure Drop (kPa)		Total Pressure Drop		Cv Value		Valve Pd	Authority %	Pipe Size (mm)		Valve Type	Actuator Model
			LPS	GPM	Coil	Fittings	kPa	psi	Required	Selected			Actual	Selected		
68	FCU/1F/30	First Floor Zone 02 Cafe L1141	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
69	FCU/1F/31	First Floor Zone 02 Cafe L1141	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
70	FCU/1F/32	First Floor Zone 02 Cafe L1141	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
71	FCU/1F/33	First Floor Zone 02 Cafe L1141	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
72	FCU/1F/34	First Floor Zone 02 Cafe L1141	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
73	FCU/1F/35	First Floor Zone 02 Cafe L1141	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
74	FCU/1F/36	First Floor Zone 03 Bridge Ceiling Void BR2	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
75	FCU/1F/37	First Floor Zone 03 Bridge Ceiling Void BR3	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
76	FCU/1F/38	First Floor Zone 03 Bridge Ceiling Void BR3	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
77	FCU/1F/39	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
78	FCU/1F/40	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
79	FCU/1F/41	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
80	FCU/1F/42	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
81	FCU/1F/43	First Floor Zone 03 Bridge Ceiling Void BR6	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
82	FCU/1F/44	First Floor Zone 03 Bridge Ceiling Void BR6	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
83	FCU/1F/45	First Floor Zone 03 Bridge Ceiling Void BR6	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
84	FCU/1F/46	First Floor Zone 03 Bridge Ceiling Void BR6	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
85	FCU/1F/47	First Floor Zone 02 Lounge L1146	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
86	FCU/1F/48	First Floor Zone 02 Lounge L1146	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
87	FCU/1F/49	First Floor Zone 02 Lounge L1146	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
88	FCU/1F/50	First Floor Zone 02 Lounge L1146	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
89	FCU/1F/51	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
90	FCU/1F/52	First Floor Zone 03 Bridge Ceiling Void BR4	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
91	FCU/1F/53	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
92	FCU/1F/54	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
93	FCU/1F/55	First Floor Zone 04 Entrance Vestibule	0.40	6.34	300.00	30.00	330.00	47.86	0.92	0.70	82.04	63.15	20.00	15.00	V5862A2039	M7410C1007
94	FCU/1F/56	First Floor External Book Drop	0.10	1.59	23.00	2.30	25.30	3.67	0.83	0.70	5.13	58.29	25.00	15.00	V5862A2039	M7410C1007
95	FCU/1F/57	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
96	FCU/1F/58	First Floor Zone 03 Bridge Ceiling Void BR5	0.10	1.59	310.00	31.00	341.00	49.46	0.23	0.20	62.81	55.95	15.00	15.00	V5862A2005	M7410C1007
97	FCU/1F/59	First Floor Zone 02 Restaurant L1150	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
98	FCU/1F/60	First Floor Zone 02 Restaurant L1150	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
99	FCU/1F/61	First Floor Zone 02 Restaurant L1150	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
100	FCU/1F/62	First Floor Zone 02 Restaurant L1150	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
101	FCU/1F/63	First Floor Zone 02 Restaurant L1150	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
102	FCU/1F/64	First Floor Zone 02 Restaurant L1150	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007

		QATAR FOUNDATION QATAR NATIONAL LIBRARY					
  		Contract No.		Document No.			
		Location EDUCATION CITY - QATAR					
		Page		Rev No.			
		419		0	1		

Equipment Name		Area Served	Valve Flow Rate		Pressure Drop (kPa)		Total Pressure Drop		Cv Value		Valve Pd	Authority %	Pipe Size (mm)		Valve Type	Actuator Model
			LPS	GPM	Coil	Fittings	kPa	psi	Required	Selected	psi		Actual	Selected		
103	FCU/1F/65	First Floor Zone 3 Bridge IDF	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
104	FCU/2F/01	Terrace Floor Zone 4 South-West 1 IDF Room	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
105	FCU/2F/02	Terrace Floor Zone 4 South-West 2 IDF Room	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
106	FCU/2F/03	Terrace Floor Zone 1 North-West IDF Room	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
107	FCU/2F/04	Terrace Floor Zone 2 North-East IDF Room	0.20	3.17	85.00	8.50	93.50	13.56	0.86	0.70	20.51	60.20	25.00	15.00	V5862A2039	M7410C1007
108	FCU/2F/05	Second Floor Zone 1 MDF Room L2009	0.36	5.71	250.00	25.00	275.00	39.89	0.90	0.70	66.45	62.49	25.00	15.00	V5862A2039	M7410C1007
109	FCU/2F/34	Second Floor Zone 03 Auditorium-High Level In Ceiling Void L2087	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
110	FCU/2F/35	Second Floor Zone 03 Auditorium-High Level In Ceiling Void L2087	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
111	FCU/2F/36	Second Floor Zone 03 Auditorium-High Level In Ceiling Void L2087	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
112	FCU/2F/37	Second Floor Zone 03 Auditorium-High Level In Ceiling Void L2087	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
113	FCU/2F/38	Second Floor Zone 03 Auditorium Booths L2087	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
114	FCU/2F/39	Second Floor Zone 03 Auditorium Booths L2087	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
115	FCU/2F/40	Second Floor Zone 03 Auditorium Booths L2087	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
116	FCU/2F/41	Second Floor Zone 02 Bridge Raised Floor (Gl E /F-4/4B) L2088	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
117	FCU/2F/42	Second Floor Zone 02 Bridge - Study Area L2088	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
118	FCU/2F/43	Second Floor Zone 03 Bridge L2088	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
119	FCU/2F/44	Second Floor Zone 03 Bridge L2088	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
120	FCU/2F/45	Second Floor Zone 03 Group Study Room L2081	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
121	FCU/2F/46	Second Floor Zone 03 Group Study Room L2076	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
122	FCU/2F/47	Second Floor Zone 03 Group Study Room L2076	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
123	FCU/2F/48	Second Floor Zone 03 Individual Study Room L2070	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
124	FCU/2F/49	Second Floor Zone 03 Individual Study Room L2070	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
125	FCU/2F/50	Second Floor Zone 03 Individual Study Room L2070	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
126	FCU/2F/51	Second Floor Zone 03 Individual Study Room L2073	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
127	FCU/2F/52	Second Floor Zone 03 Individual Study Room L2073	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
128	FCU/2F/53	Second Floor Zone 03 Individual Study Room L2073	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
129	FCU/2F/54	Second Floor Zone 03 Bridge Raised Floor (Gl E /F-6/6B) L2088	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
130	FCU/2F/55	Second Floor Zone 03 Bridge Raised Floor (Gl E /F-6/6B) L2088	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
131	FCU/2F/56	Terrace Floor Zone 02 North East Core T1021	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
132	FCU/2F/57	Terrace Floor Zone 02 North East Core T1021	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
133	FCU/2F/58	Terrace Floor Zone 02 North East Core T1021	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
134	FCU/2F/59	Terrace Floor Zone 02 North East Core T1021	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
135	FCU/2F/60	Terrace Floor Zone 01 North West Core T1020	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
136	FCU/2F/61	Terrace Floor Zone 01 North West Core T1020	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007
137	FCU/2F/62	Terrace Floor Zone 01 North West Core T1020	0.10	1.59	256.00	25.60	281.60	40.84	0.25	0.20	62.81	60.60	20.00	15.00	V5862A2005	M7410C1007